Matrix discrepancy and optimization

The sharp Lovász-theta constant for dense random graphs

Difficulty: challenging **Importance:** broadly interesting

Status: source-backed open problem; no resolution found in the bounded literature check.

Let G_n be the simple random graph on n labelled vertices in which each unordered pair is an edge independently with probability $1/2$. For a graph G, define its Lovász number by

$$\vartheta(G) = \max\{\langle J, X\rangle : X \in \mathbb{R}^{n\times n},\ X = X^T \succeq 0,\ \operatorname{tr} X = 1,\ X_{ij} = 0 \text{ for } \{i,j\} \in E(G)\},$$

where J is the all-ones matrix and $\langle J, X\rangle = \sum_{i,j} X_{ij}$.

Conjecture. $\lim_{n\to\infty} \mathbb{E}\,\vartheta(G_n)/\sqrt{n} = 1$.

This asks for the leading constant of a random semidefinite matrix optimization problem. It is an expectation statement; convergence in probability is not substituted for it.

References

1. A. S. Bandeira, D. Dmitriev, K. Lucca, P. Nizić-Nikolac, and A. Rödder, *Randomstrasse101: Open Problems of 2025*, Entry 9, Conjecture 17 and the preceding SDP definition. Archived paper.

Status check — 2026-09-08

The archived 2026 paper retains the conjecture. Targeted searches for the Lovász number of dense Erdős–Rényi graphs and a sharp leading constant through September 2026 found no resolution. The adjacent circulant problem uses a different random matrix ensemble and is recorded separately.